

The gender-dependent development of moral exclusiveness

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Statement of author contributions

The studies in this paper were conceptualized and designed by all authors. Data was collected by LM and AC. The first draft was written by LM, with feedback and editing from AC and NF.

Abstract

Men are more morally exclusive than women. When and how does this difference emerge in human development? Across four samples spanning the ages of 4 to 80 (total $N = 1,842$), we measure people's moral exclusiveness toward human outgroups and non-human animals in both attitudes and related behaviour. Replicating existing findings, we demonstrate that adult men are more morally exclusive than women in both attitude and behaviour. Crucially, we identify adolescence (13–18 years) as the key window for development: older male adolescents are more morally exclusive than younger male adolescents, while there is no such difference for female adolescents. Linked to this is an increase in social dominance orientation with age in males. We further find the onset of gendered moral exclusiveness: while males are already more morally exclusive than females at the age of 6 and such gender differences persist throughout childhood and in adulthood, they are not yet evident in younger children.

Statement of limitations

First, our studies were conducted only in a WEIRD cultural context, so we cannot generalise our findings to other populations. Second, while the adult version of the speciesism scale has been subjected to all stages of scale development (Caviola et al., 2019), our adapted scale for younger age groups has not undergone separate psychometric testing. Third, our work is cross-sectional rather than longitudinal, and as such we cannot entirely rule out the possibility of cohort effects. Forth, to not overburden young children with experimental materials, we have measured moral exclusiveness with one scale in the child samples (compared to several amongst adolescents and adults), which limits the ability to draw general conclusions.

Key words: Moral exclusiveness, Outgroup Warmth, Speciesism, Social Dominance Orientation, Development

In recent political elections around the globe, (young) men have been increasingly likely to support right wing or populist political parties as compared to women in the same age bracket (YouGov, 2026; CIRCLE, 2024; Jung, 2024; Milosav et al., 2026). These political causes that appeal to many young men are united by a focus on prioritising one's national ingroup and in many cases downgrading immigrant groups. In other words, they share a perspective on dominance, hierarchies, and who ought to be morally included. From a psychological perspective, gender-dependent differences in moral exclusiveness and prejudice between adult men and women are well documented. Men have been shown to be more authoritarian (Brandt & Henry, 2012), to score higher in hostile sexism (Bareket & Fiske, 2023), report greater homophobia (Herek, 1988), derogate animals more (Caviola et al., 2019; Salmen & Dhont, 2023), and to hold less impartial moral views (Kahane et al., 2018). In contrast, women are more morally inclusive towards human and non-human groups (Crimston et al., 2016), extending as far as seeing entire ecosystems as worthy of greater moral concern than men do (Zelezny et al., 2000) and being more likely to restrict their consumption of non-human animals (Rosenfeld, 2023). Where does this stark difference between men and women in such a crucial domain of social life come from? *When* and *how* do men and women, or boys and girls, begin to diverge in terms of moral exclusiveness?

To answer these questions, we examine markers of moral exclusiveness amongst male and female participants ranging from the age of 4-years-old to 80-years-old, focusing on different targets including human outgroups (measured amongst adolescents and adults) and non-human animals (measured amongst younger children, adolescents, and adults). First, we replicate existing findings on adult gender differences in moral exclusiveness towards human outgroups and non-human animals (Caviola et al., 2019; Dhont et al., 2016) in a large sample representative of the UK population, showing that men are more morally exclusive than women in both judgment measures and a relevant behaviour (i.e. meat-eating). We also show

that increased moral exclusiveness in men's judgments is associated with higher social dominance orientation (i.e., a higher preference for social hierarchies). Then, testing the importance of adolescence (13 – 18-year-olds) as a key developmental period, we examine gender differences in moral exclusiveness again in judgements and behavioural correspondence, finding that older adolescent boys are more morally exclusive than younger adolescent boys – which is not the case for adolescent girls. Again, these judgments are associated with social dominance orientation. Finally, we examine two younger samples, documenting that the onset of gender differences in moral exclusiveness lies in middle childhood, already at the age of 6, while in early childhood, boys and girls make equally morally inclusive judgments.

Moral Exclusiveness and the Preference for Hierarchies

Children are generally more morally inclusive than adults (Caviola et al., 2025; Elenbaas & Killen, 2016a, 2016b; Marshall et al., 2025; McGuire et al., 2023, 2026; Rizzo et al., 2016; Wilks et al., 2021). We propose that to understand how morally inclusive children develop into more morally *exclusive* men and *inclusive* women, it is key to consider the role of social hierarchies. People have been shown to vary in their endorsement for the degree to which relations between groups ought to be organised hierarchically. Studies of this preference for hierarchies have primarily been based on Social Dominance Theory and accordingly relied upon measures of *social dominance orientation* (Pratto et al., 1994; Sidanius & Pratto, 1999), which reflects individual-level preferences for the dominance of “superior” over “inferior” groups. The key manifestation of such socially dominant thinking is the expression of moral exclusiveness. For example, social dominance orientation has been shown to relate to racism (Duriez & Soenens, 2009), sexism (Austin & Jackson, 2019), and attitudes towards animals (Caviola et al., 2019). Such findings align with the concept of generalized prejudice and the idea that these different expressions of prejudice all share an

underlying psychology (Bergh & Brandt, 2023; Everett et al., 2019). Accordingly, greater racism, sexism, homophobia, and downgrading of non-human animals (speciesism) are expressed at least in part due to greater endorsement of social dominance orientation (Dhont et al., 2016).

Regarding gender differences, in support of Social Dominance Theory (Sidanius & Pratto, 1999), there is extensive evidence documenting that men score higher on social dominance orientation measures than women (Batalha et al., 2011; Foels & Reid, 2010; Lee et al., 2011; Sidanius et al., 1994). The argument has been made that such gender differences in the preference for social hierarchies are in part a residue of our evolutionary past (McDonald et al., 2011; Pratto et al., 2006; Sidanius et al., 2024; Sidanius & Pratto, 1999). Limited access to resources of all kinds spurred competition between individuals and groups, and social hierarchies – the status of an individual or group – determined who received access to these resources. It has long been argued that different selection pressures on males and females made males more prone to intra- and inter-group competition than females (Emlen & Oring, 1977) and in turn, to show preferences for the social hierarchies which emerge from this competition.

As such, gender differences in prejudices against other groups can be interpreted as exemplars of a more general preference for hierarchies and dominance (Sidanius & Pratto, 1999). If a psychology of hierarchies can at least in part explain why adult men and women show differing levels of moral exclusiveness, the open question remains as to *when* gendered differences in this psychology may emerge.

The Key Role of Adolescence

We argue that in the development of moral exclusiveness and social dominance orientation, adolescence is the key period where differences between males and females should solidify. Adolescence is a particularly important period (Sawyer et al., 2018), during

which humans achieve a wide array of social-developmental milestones and navigate significant changes in the social brain. Importantly for our argument concerning social hierarchies, some evidence suggests that brain regions implicated in the processing of status are developing during adolescence (Ansari & Dhital, 2006; Burnett et al., 2011; Gogtay et al., 2004). During adolescence, group life increases in importance (Miklikowska et al., 2019; Reitz et al., 2014), and there is first evidence of more morally exclusive judgments as compared to childhood. For example, adolescents are more morally exclusive towards animals (McGuire et al., 2025) and selectively allocate resources towards in-group members (Güroğlu et al., 2014).

Importantly, social dominance orientation has been shown to relate to some examples of moral exclusiveness during adolescence (Kudrnáč, 2017; Poteat & Anderson, 2012). During this period there are also important gender differences in a proclivity towards competition. Adolescent boys engage in more competing to win than adolescent girls (Hibbard & Buhrmester, 2010), are more involved in competitive sports from pre-adolescence onwards (Mathur & Berndt, 2006; Zarbatany et al., 2000), and more likely to display direct aggression (Card et al., 2008). Taken together, it appears the adolescent brain is prepared to organise around status hierarchies, adolescents in some cases are more morally exclusive, and adolescent boys are particularly motivated towards competition. Given this, we expect that adolescent boys are likely to express greater moral exclusiveness judgments and behaviour than adolescent girls and do so at least in part due to an emerging focus on hierarchies.

The Onset of Gendered Moral Exclusiveness

Long before adolescence, children use gender as an important category for understanding the social world (Liben & Bigler, 2002). For humans, gender has primacy as a social category (Martin & Ruble, 2010; Ruble et al., 2006) – already very early in life, infants

distinguish between male and female faces (Quinn et al., 2002). This is followed, in the first two years of life, with children learning terms like ‘boys’ and ‘girls’ and subsequently making gender-typed toy choices (Campbell et al., 2002; Zosuls et al., 2009). Gender is not only helpful for categorising the social world but carries significant meaning for children.

Throughout childhood, humans come to hold beliefs about what kinds of possessions, appearance characteristics, and activities are appropriate for members of gender groups (Leinbach et al., 1997; Weinraub et al., 1984). These norms and roles are also related to ideas about who has status, with children more likely to attribute ‘brilliance’ to boys than to girls (Bian et al., 2017). In turn, children endorse gender differences regarding how we ought to behave towards others, with children over 6-years-old associating more communal values with girls than boys – and boys who endorse the association between girls and communality showing less interest in communal activities themselves (Block et al., 2025). Further recent evidence has emphasised that these ideas are socially learned, with 5 – 6 -year-old boys and girls showing an equal preference for competitive games, whereas 13 – 14-year-old boys demonstrate a greater preference for competitive games than girls (Asoy & Gokdemir, 2025).

Crucially, already in middle childhood, that is between the ages of 7-years-old and 10-years-old, children begin to understand society’s ideas about masculinity and what it means to “be a man” (Sinno & Killen, 2009). One such gender role that may be communicated to children is the idea that men ought to act in a more morally exclusive manner. Such an idea may lead to early differences between boys and girls, and to set the stage for a stronger divergence between male and female adolescents during puberty. We note that it may be possible to find a point where gender differences in moral exclusiveness are not yet apparent, due to the socially learned aspect of this and related behaviours. Therefore, while we argue that adolescence is likely the key period for a divergence between males and females’ moral exclusiveness, we also stress the importance of exposure to society’s gender roles and norms,

which may manifest in early gender differences even before the onset of puberty, though the age of onset of such differences is unknown. Therefore, here we also ask at what early age indications of gender differences in moral exclusiveness become apparent.

The Current Project

Taken together, we compare moral exclusiveness of male and female participants in their judgements and behavioural markers, using groups of adults, adolescents, and younger children. Amongst adults and adolescents, we ask (a) whether there are gender differences in moral exclusiveness judgments toward human outgroups and animals, (b) whether there is a gender difference in a morally exclusive behaviour, and (c) to what extent moral exclusiveness judgments are related to increased social dominance orientation. Amongst children we ask (d) whether there are gender differences in moral exclusiveness judgments towards animals.

Methods

Transparency and Openness

The sample, methods, and linear models for the adult and early childhood samples were pre-registered on AsPredicted (adults: <https://aspredicted.org/fy4w-23fh.pdf>, early childhood: <https://aspredicted.org/yn2c-3syv.pdf>). Additional pre-registered analyses (of age and gender effects on 4N endorsement) can be found in Section 3 of the supplemental information. The chi-square analysis of dietary restrictions in the adult sample should be considered exploratory. The middle childhood and adolescent studies were not pre-registered. The materials, data, and analysis of all four studies can be accessed on ResearchBox (https://researchbox.org/3942&PEER_REVIEW_passcode=UOQJHV). The study received ethical clearance from the [BLINDED] ethics committee. See supplemental information Tables 1 and 2 for full pre-registration information including deviations from pre-registration. For all samples we report sensitivity analysis of the smallest detectable sample size.

Participants and Procedure

The study consists of four samples of UK participants, each representing a different age group, with a total $n = 1,842$.

Adults

The initial sample included 800 participants recruited from Prolific Academic and was representative in terms of age, gender, and ethnicity of the UK population. The survey took a median of 7 minutes and 10 seconds to complete, and participants were compensated £1.50 for their time. All participants gave informed consent to participate. As gender was a central variable in the analysis, participants who did not report their gender ($n = 4$), or reported their gender as non-binary ($n = 7$) were omitted from the analysis. This left a final sample of 789 participants (M age = 46.30, $SD = 15.50$, min = 18 years, max = 80 years) with 381 male adults and 408 female adults. A sensitivity analysis based on a linear model with 789 participants, 4 covariates, an alpha level of 0.05 and power of 0.8 indicated a minimum detectable effect of $f^2 = 0.015$ or $R^2 = 0.015$.

Amongst these participants, the majority identified as White (including British, Irish, or Irish Traveller; $n = 676$). The sample also included Asian (including Indian, Pakistani, Bangladeshi; $n = 59$), Black (including Caribbean, African, Black British; $n = 26$), Mixed or multiple ethnic groups (including White and Black Caribbean, White and Asian; $n = 15$) and other ethnic groups ($n = 13$).

Most of the sample were omnivorous ($n = 629$) but also included vegetarians ($n = 56$), vegans ($n = 16$), pescatarians ($n = 24$), flexitarians ($n = 44$), and participants who held another diet ($n = 13$). Seven participants did not report their diet and were not included in the analysis of morally exclusive behaviour.

Adolescents

The initial sample included 529 participants from the UK, who were recruited via schools and whose parents gave informed consent for their children to participate. Participants who did not report their gender ($n = 15$) or reported their gender as other ($n = 23$) or non-binary ($n = 13$) were omitted from the analysis. This left a final sample of 478 participants (M age = 15.19, $SD = 1.45$, min = 13 years, max = 18 years) with 233 male adolescents and 245 female adolescents. A sensitivity analysis based on a linear model with 478 participants, 4 covariates, an alpha level of 0.05 and power of 0.8 indicated a minimum detectable effect of $f^2 = 0.025$ or $R^2 = 0.025$.

Amongst these participants, the majority identified as White (including British, Irish, or Irish Traveller; $n = 404$). The sample also included Asian (including Indian, Pakistani, Bangladeshi; $n = 27$), Black (including Caribbean, African, Black British; $n = 5$), Mixed or multiple ethnic groups (including White and Black Caribbean, White and Asian; $n = 24$) and other ethnic groups (including Arab; $n = 14$), as well as four participants who did not report their ethnicity.

Most of the sample were omnivorous ($n = 411$) but also included vegetarians ($n = 10$), vegans ($n = 4$), pescatarians ($n = 10$), flexitarians ($n = 13$), and participants who held another diet ($n = 17$). Thirteen participants did not report their diet and were not included in the analysis of morally exclusive behaviour.

Middle Childhood

The sample included 379 children from the UK (M age = 8.51, $SD = 1.36$, min = 6 years, max = 11 years) with 192 male children and 193 female children, who were recruited via schools and whose parents gave informed consent for their children to participate. A sensitivity analysis based on a linear model with 379 participants, 4 covariates, an alpha level of 0.05 and power of 0.8 indicated a minimum detectable effect of $f^2 = 0.03$ or $R^2 = 0.03$.

The majority of participants identified their ethnicity as White British ($n = 111$) and the sample included Mixed Race/Dual Heritage ($n = 20$), South Asian British (including Bengali, Indian, and Pakistani British participants, $n = 32$), Black British ($n = 8$), Chinese British ($n = 15$), other ethnic groups ($n = 30$) and participants who did not report their ethnicity ($n = 53$).

Most of the sample were omnivorous ($n = 305$) but also included vegetarians ($n = 34$), vegans ($n = 6$), pescatarians ($n = 3$) and flexitarians ($n = 2$).

Early Childhood

The sample comprised 196 children (M age = 5.14, $SD = 0.77$, min = 4 years, max = 6 years) including 104 female children and 92 male children. Participants were visitors to a zoo who were approached at the beginning of their visit (i.e., before seeing any animals) by an experimenter who offered them the chance to take part in a short survey in exchange for a sticker. Parents provided consent before an experimenter talked their child through the survey. A sensitivity analysis based on a linear model with 196 participants, 3 covariates, an alpha level of 0.05 and power of 0.8 indicated a minimum detectable effect of $f^2 = 0.06$ or $R^2 = 0.05$.

Most participants were White British ($n = 188$) and the sample also included participants who were Black British ($n = 1$), Mixed Race/Dual Heritage ($n = 4$) and other ethnic groups ($n = 3$).

Most of the sample were omnivorous ($n = 178$) but also included vegetarian or vegan participants ($n = 10$) and participants who did not report their diet ($n = 8$).

Procedure

All participants completed a survey electronically via Qualtrics. The adolescent and the middle childhood surveys contained some measures that don't speak to the questions at hand, so are not reported here (adolescent survey: participants' factory farming knowledge,

need for belonging, perceived peer norms, and school inclusion; middle childhood survey: food neophobia and preferences for different fruits and vegetables).

Measures

Moral Exclusiveness Toward Human Outgroups

As the measure of moral exclusiveness towards human outgroups, we used an attitude temperature gauge to assess adults' and adolescents' feelings of warmth towards outgroup members (Dhont et al., 2016). Participants were asked to indicate on a sliding scale from 0 (*very unfavourable*) to 100 (*extremely favourable*) how warmly they felt towards six groups (Immigrants, Vegans, Ethnic Minorities, Eastern Europeans, Farmers, British people). Immigrants, ethnic minorities, and Eastern Europeans were chosen as salient outgroups in the UK context and a mean of these three warmth items was created for analysis (adults, $\alpha = .90$; adolescents, $\alpha = .87$). To assess differences in moral exclusiveness rather than outgroup warmth, we reverse coded each participant's response.

Moral Exclusiveness Toward Non-Human Animals

As a measure of moral exclusiveness towards non-human animals we used the six-item speciesism scale (Caviola et al., 2019). For middle childhood and adolescent participants, the language adapted version of this scale (McGuire et al., 2023) was used. This scale (adults, $\alpha = .80$; adolescents, $\alpha = .83$; middle childhood, $\alpha = .65$; early childhood, $\alpha = .59$) assessed participants' agreement with statements regarding humans' intrinsic moral worth as compared to non-human animals (e.g., "it is okay to test new medicines on animals that we wouldn't test on humans; 1 = *strongly disagree*, 7 = *strongly agree*).

To create a version of the scale appropriate for use with 4-year-olds in the early childhood sample, we rephrased each statement of the adapted child scale into a question (e.g., "Animals are worth less than people" became "are animals less important than people?") and first presented three response options: a sad face, a neutral face, and a smiling

face. Participants were asked to indicate their agreement using a smiling face, disagreement using the sad face, or if they weren't sure, to pick the neutral face. If participants picked the sad face or the happy face, they were then shown three additional sad or happy faces which increased in magnitude of expression so they could further specify their answer. If participants picked the neutral face they moved onto the next question. All versions of the speciesism scale can be found in the supplemental information (see section 4).

Morally Exclusive Behavior

As a behavioural correspondence for moral exclusiveness, we focused on non-human animals and assessed whether participants ate animals or restricted their diet to any degree. Specifically, morally exclusive behaviour was defined as not limiting amount of meat consumption (i.e., participant did not indicate they followed a vegetarian, vegan, pescatarian, or flexitarian diet). We chose eating behaviour for several reasons. First, compared to other forms of morally exclusive behaviour it is highly prevalent and socially acceptable, so less prone to floor effects or response bias. Second, it can be measured in adulthood and adolescence, as the question is comprehensible across these ages, and adolescents at least have some control of their diet in general. Third, it was plausible to assume there would be a gender-dependent aspect in adults (Hopwood et al., 2024), which allowed us to assess this in the younger samples.

Preference for Hierarchies

For adult and adolescent participants, the four-item short social dominance orientation scale (Pratto et al., 2014) was used to measure participants' endorsement of socially dominant thinking (e.g., "We should not push for group equality", 1 = *very strongly against this idea*, 10 = *very strongly support this idea*; adults, $\alpha = .81$; adolescents, $\alpha = .67$).

Results

Gender differences in moral exclusiveness amongst adults

All analyses were conducted using R (Team, 2024) and RStudio (team, 2025). See Supplemental Tables 3 through 10 for the full results of all linear models.

In a linear model analysis, we assessed whether the association between gender and different forms of moral exclusiveness differed when controlling for social dominance orientation. Male adults, compared to female adults, reported more morally exclusive judgements (see Figure 1), both towards human outgroups ($B = -1.65$, $SE = 0.73$, $LLCI = -3.08$, $ULCI = -0.21$, $t(768) = -2.25$, $p = .02$, $d = -0.08$) and animals ($B = -0.29$, $SE = 0.04$, $LLCI = -0.36$, $ULCI = -0.22$, $t(774) = -7.92$, $p < .001$, $d = -0.28$).

We conducted chi-square analysis to understand whether gender was related to dietary restrictions (e.g., veg*n, flexitarian, pescatarian) amongst this sample. Gender was related to this morally exclusive behaviour ($X^2(1) = 14.71$, $p < .001$), with 86% of male adults engaging in a morally exclusive behaviour (i.e., not restricting their diet to exclude some or all animal products) compared to 75% of female adults.

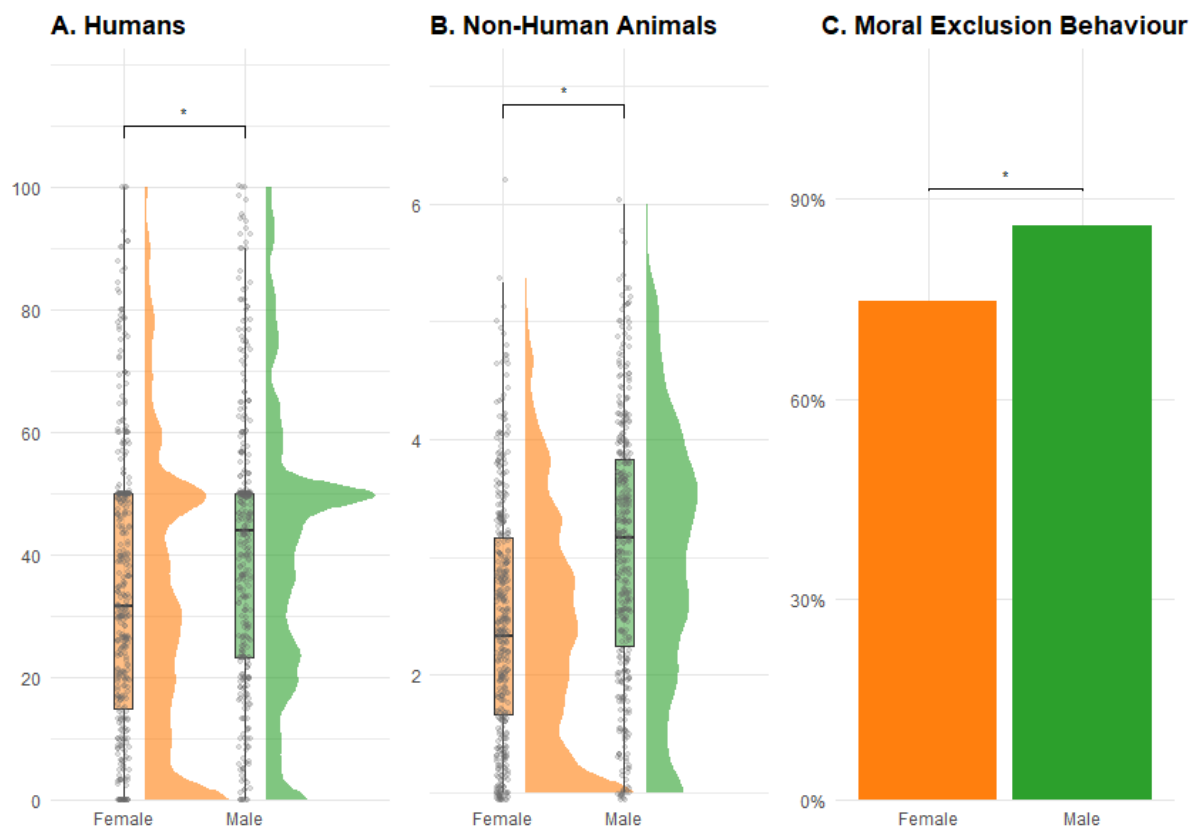
Male adults, compared to female adults, scored higher in social dominance orientation ($R^2 = .04$, $F(3, 780) = 12.47$, $p < .001$; $B = -0.32$, $SE = 0.06$, $LLCI = -0.44$, $ULCI = -0.20$, $t(745) = -5.00$, $p < .001$, $d = -0.18$). Age was also a predictor of social dominance orientation, with older adults scoring higher than younger adults ($B = 0.01$, $SE = 0.004$, $LLCI = 0.006$, $ULCI = 0.02$). Age and gender did not interact.

To further understand the relationship between gender, speciesism and out-group attitudes amongst adult participants, we conducted partial correlation analysis accounting for the role of social dominance orientation. The association between gender and morally exclusive judgements towards human outgroups ($r = 0.16$, $t(783) = 8.12$, $p < .001$, $LLCI = 0.10$, $ULCI = 0.23$) was weakened when accounting for social dominance orientation ($r = 0.10$, $t(782) = 2.69$, $p = .007$, $LLCI = 0.03$, $ULCI = 0.16$). Likewise, the association between gender and moral exclusiveness towards animals ($r = 0.28$, $t(783) = 8.12$, $p < .001$, $LLCI =$

0.21, $ULCI = 0.34$) was weakened when accounting for social dominance orientation ($r = 0.24$, $t(782)$, $p < .001$, $LLCI = 0.18$, $ULCI = 0.31$).

Figure 1

Moral exclusiveness (morally exclusive judgements about a. other humans, b. non-human animals, and c. percentage of participants showing morally exclusive behaviour) as a function of participant gender in adults aged between 18 and 80. Raincloud plots reflect mean per group with 95% confidence intervals.

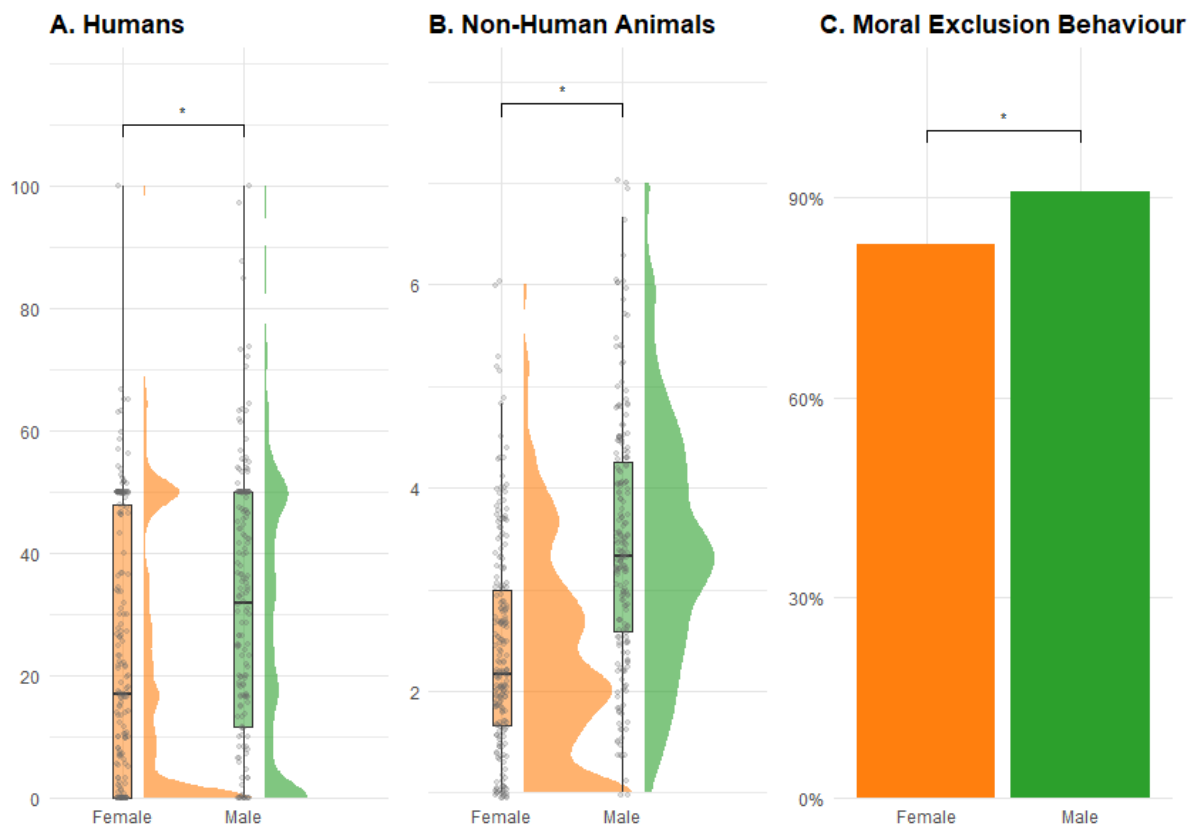


The development of gendered moral exclusiveness during adolescence

Male adolescents, compared to female adolescents, reported more exclusive judgments (see Figure 2), regarding human outgroups ($B = -3.35$, $SE = 1.15$, $LLCI = -5.61$, $ULCI = -1.09$, $t(349) = -2.92$, $p = .004$, $d = -0.16$) and animals ($B = -0.50$, $SE = 0.06$, $LLCI = -0.61$, $ULCI = -0.38$, $t(399) = -8.75$, $p < .001$, $d = -0.44$). Again, gender was related to morally exclusive behaviour ($\chi^2(1) = 5.85$, $p = .02$), with 91% of male adolescents engaging in a morally exclusive behaviour compared to 83% of female adolescents.

Figure 2

Moral exclusiveness (morally exclusive judgements about a. other humans, b. non-human animals, and c. percentage of participants showing morally exclusive behaviour) as a function of participant gender in adolescents aged between 13 and 18. Raincloud plots reflect mean per group with 95% confidence intervals.



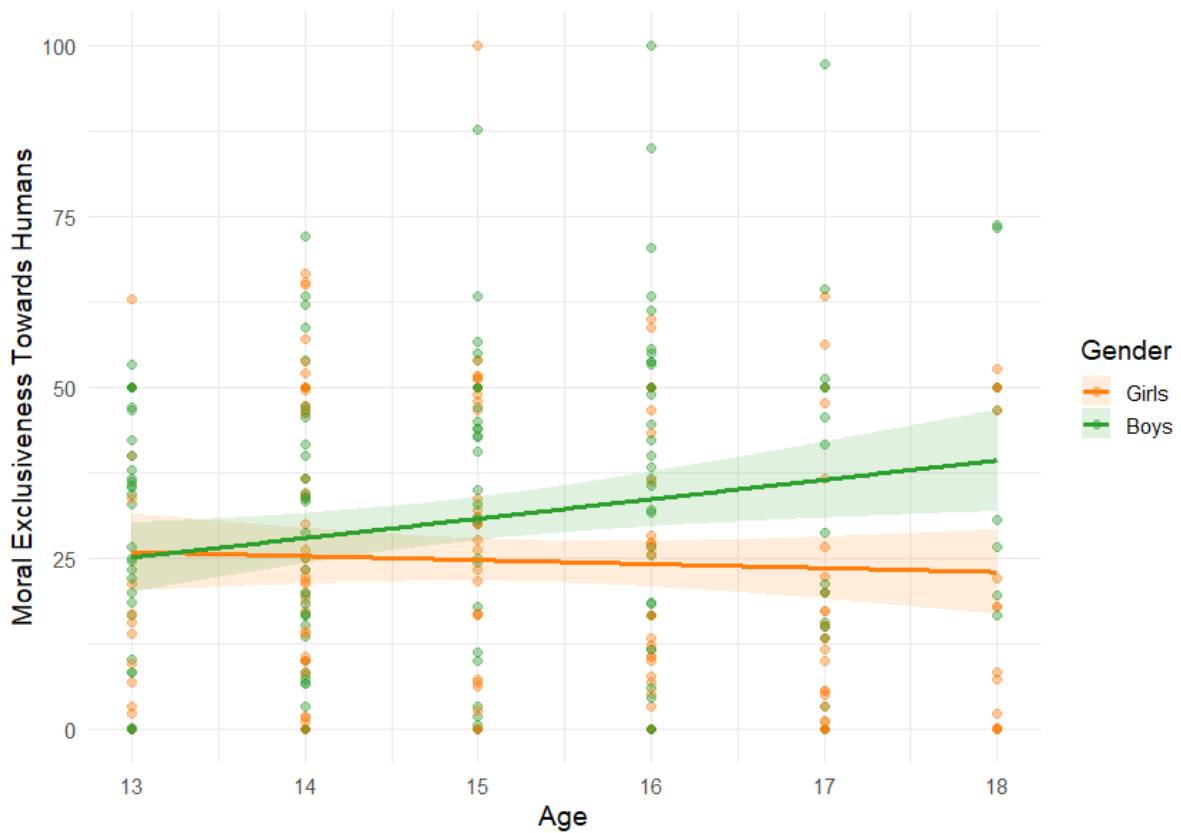
During adolescence the developmental trajectory of moral exclusiveness diverged for male and female adolescents, especially concerning human outgroups. The effect of gender on moral exclusiveness towards human outgroups was qualified by an interaction (see Figure 3) between gender and age ($B = -1.71$, $SE = 0.75$, $LLCI = -3.19$, $ULCI = -0.24$, $t(349) = -2.28$, $p = .023$, $d = -0.12$). Older male adolescents were likely to be more morally exclusive toward human outgroups than younger male adolescents ($B = 2.84$, $SE = 1.09$, $LLCI = 0.70$, $ULCI = 4.99$, $p = 0.01$) whereas for female adolescents there was not a significant relationship

between age and moral exclusiveness ($B = -0.58$, $SE = 1.03$, $LLCI = -2.61$, $ULCI = 1.45$, $p = 0.57$).

The effect of gender on moral exclusiveness toward animals was also qualified by an interaction between gender and age ($B = -0.09$, $SE = 0.04$, $LLCI = -0.17$, $ULCI = -0.02$, $t(399) = -2.41$, $p = .016$, $d = -0.12$). Simple slopes analyses suggested that while there was a trend of greater moral exclusiveness amongst older male adolescents ($B = 0.10$, $SE = 0.06$, $LLCI = -0.01$, $ULCI = 0.20$, $p = .09$) and a trend of lower moral exclusiveness amongst older female adolescents ($B = -0.09$, $SE = 0.05$, $LLCI = -0.19$, $ULCI = 0.01$, $p = .09$), in neither case was this significant.

Figure 3

Adolescents' moral exclusiveness towards humans as a function of age and gender



Again, we assessed whether the association between gender and different forms of moral exclusiveness differed when accounting for social dominance orientation. Male adolescents, compared to female adolescents, were likely to score higher in social dominance orientation ($R^2 = .05$, $F(3, 387) = 7.70$, $p < .001$; $B = -0.39$, $SE = 0.09$, $LLCI = -0.57$, $ULCI = -0.21$, $t(387) = -4.25$, $p < .001$, $d = -0.21$). There was no effect of age, nor an interaction between age and gender. Among adolescents, the association between gender and moral exclusiveness towards human outgroups ($r = 0.18$, $t(345) = 3.46$, $p = .0006$, $LLCI = 0.08$, $ULCI = 0.28$) was weakened when accounting for social dominance orientation ($r = 0.12$, $t(344) = 2.32$, $p = .021$, $LLCI = 0.02$, $ULCI = 0.23$). Likewise, the association between gender and moral exclusiveness towards animals ($r = 0.39$, $t(345) = 7.77$, $p < .001$, $LLCI = 0.29$, $ULCI = 0.47$) was weakened when accounting for social dominance orientation ($r = 0.34$, $t(344)$, $p < .001$, $LLCI = 0.24$, $ULCI = 0.43$).

Gender differences in moral exclusiveness during middle and early childhood

In middle childhood (mean age = 8.5-years-old), male children, compared to female children, were likely to report greater moral exclusiveness judgements regarding animals ($B = -0.21$, $SE = .05$, $LLCI = -0.31$, $ULCI = -0.11$, $t(337) = -4.16$, $p < .001$, $d = -0.22$; see Figure 4). The gender difference in moral exclusiveness was apparent at the youngest age in the middle childhood sample (6 – 7-years-old), $t(76) = -2.71$, $p = .008$, $LLCI = -1.09$, $ULCI = -0.17$, $d = -.58$.

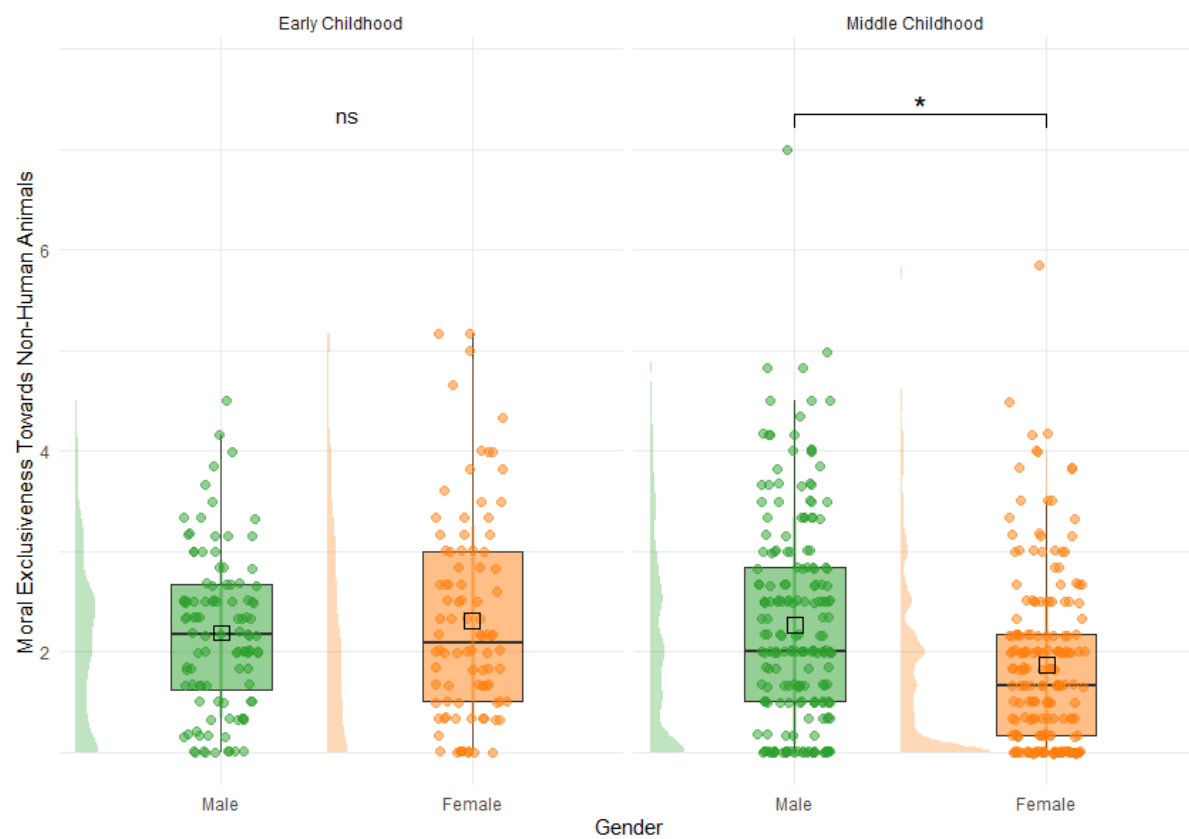
There was no relation between gender and morally exclusive behaviours; $\chi^2(1) = 0.005$, $p = .942$, with 88% of male children engaging in a morally exclusive behaviour, compared to 87% of female children. It is worth noting that while children exist that chose a vegetarian diet while their parents do not (Hussar & Harris, 2010), most children eat what is prepared by their parents, and as such choice of dietary restriction is likely a less appropriate

behavioural marker of moral exclusiveness in this age group as compared to the older participants.

To probe further, we assessed the effect of gender on moral exclusiveness toward animals amongst a separate early childhood sample (mean age = 5.14-years-old). Here, there was notably no significant relationship between gender and moral exclusiveness ($B = -0.04$, $SE = 0.07$, $LLCI = -0.17$, $ULCI = 0.09$, $t(183) = -0.59$, $p = .559$, $d = -0.05$; see Figure 4).

Figure 4

Moral exclusiveness towards Non-Human Animals as a function of participant gender in early (Mean Age = 5.14 years) and middle (Mean Age = 8.5 years) childhood. Raincloud plots reflect mean per group with 95% confidence intervals.



Discussion

There is a growing social and political divide between adult men and women: men are more morally exclusive towards other individuals than women are (Brandt & Henry, 2012; Caviola et al., 2019; Herek, 1988; Rosenfeld, 2023). We sought to understand *when* men and women, or boys and girls, might start to diverge in their moral exclusiveness judgments and related behaviour, and to shed light on a potential explanation for this gender difference. Our key findings are 1) that already at the age of 6-years – but not earlier in childhood – male children are more morally exclusive than female children, 2) that adolescence is the key period for the development of increased moral exclusiveness in males, and 3) that in adolescence, the higher moral exclusiveness of males is associated with a higher general preference for social hierarchies.

We measured moral exclusiveness in judgements about human outgroups (outgroup warmth) and non-human animals (speciesism), as well as a behavioural proxy for morally inclusive behaviour (dietary restriction of meat consumption). It has been well-established that moral exclusiveness towards other humans is linked to moral exclusiveness towards animals (Caviola et al., 2019; Dhont et al., 2016; Everett et al., 2019) and that moral exclusiveness towards animals makes restricting one's diet to exclude animals less likely (Caviola et al., 2019; McGuire et al., 2023; Rosenfeld, 2023). Replicating earlier research (Caviola et al., 2019; Dhont et al., 2016) we found that gender is consistently related to moral exclusiveness amongst adults. Male adults, compared to female adults, are more morally exclusive towards human outgroups and non-human animals, as well as more likely to engage in a morally exclusive behaviour in the form of not restricting eating animals in any way. Further, we find evidence of an association between moral exclusiveness and a higher preference for hierarchies as measured by a higher social dominance orientation.

Adolescence is Key for Gendered Moral Exclusiveness

Male adolescents were also more morally exclusive than female adolescents on all three measures. Crucially, across the adolescent age range, there was a trend towards greater moral exclusiveness amongst older males compared to younger males, which was not the case among female adolescents. Adolescence is a central period for social and cognitive development, where adolescent males are particularly sensitive to the risk of loss of status (London et al., 2007; Marston et al., 2010). Adolescents are aware of the social norms and roles they are expected to fulfil, particularly those that fall along gender lines (Dotti Sani & Quaranta, 2017) and indeed those related to the treatment of other beings (Carter et al., 2025). With the increasing importance of normative processes in adolescence, male adolescents may be more likely to align with expectations regarding status maintenance through morally exclusive attitudes and behaviours. For example, the association between meat consumption and perceived masculinity is well documented (Rosenfeld, 2023; Rozin et al., 2012; Salmen et al., 2025). During this period adolescent males increasingly align themselves with the attitudes and behaviours that uphold such views. Again, the relation between gender and moral exclusiveness was weakened when accounting for social dominance orientation. Interestingly, these findings align with evolutionary accounts regarding humans (McDonald et al., 2011; Sidanius et al., 2024) and evidence from other social living species (Sandel et al., 2017), particularly amongst mammals where adolescence is a period of increased status concern and competition (Anestis, 2006; Bloomsmith et al., 1994; Enigk et al., 2021; Holekamp & Smale, 1993). Whether evolutionarily prepared or socially constructed, the adolescent brain organises around hierarchical structures, which have increasing importance for males.

Possible Origins of Early Gendered Moral Exclusiveness

Given our analyses reveal the importance of adolescence for gender differences in moral exclusiveness, a further question arises regarding the early onset of these gender

differences. Therefore, among two samples of children, we documented the onset of gender differences in moral exclusiveness. Specifically, 6-year-old male children were more morally exclusive than 6-year-old female children. When asked about the worth of non-human animals, 6-year-old boys were more likely to endorse the idea that humans have greater moral value than non-human animals. This points to the key role of social construction in gender and moral exclusiveness. Already children endorse ideas that align with the expectations of adult men and women, such as their parents' own endorsement of hierarchies (Guidetti et al., 2021). An open question for future research is to better understand why gender differences emerge more strongly in middle childhood (i.e., between 6 and 12-years-old), despite gender socialization occurring from birth. Some differences in social dominance orientation are related to the way in which mothers talk to their children about outgroup members (Kong et al., 2025) and recent evidence has documented that self-reported SDO in children aged 5 – 12-years-old is associated with lower acceptance of out-group suffering (Lei et al., 2025), suggesting again that early differences in the psychology of hierarchies emerge *before* the onset of puberty. Examining the emergence of preference for hierarchy and how this is in part socialised differently between boys and girls in early to middle childhood will be a valuable step.

Pathways to Gendered Moral Exclusiveness

Together, our findings support (a) the idea that adolescence is a key developmental period for moral exclusiveness and preference for hierarchies, as well as (b) the proposition that human children are already being prepared for their social roles, beginning to differentiate their behaviour based on gender by 6-years-old. This evidence points to two related pathways through which adult gender differences in moral exclusiveness may begin to occur. First, a divergence in moral exclusiveness judgments between male and female adolescents is consistent with arguments from evolutionary psychology that during

adolescence, male humans benefit from engagement in status maintaining behaviours. The expression of greater moral exclusiveness towards other human and non-human animal groups, as well as greater engagement in relevant behaviours are possible signals or consequences of this increasing status concern during adolescence. At the same time, 6-year-old male children already demonstrated greater moral exclusiveness than female children of the same age. Such differences well before the onset of puberty allow for the speculation that gendered moral exclusiveness is at least in part socially constructed. Indeed, humans did not always live along unequal gendered lines, with this hierarchical structure only emerging because of humans' transition from nomadic foraging groups to sedentary agricultural societies (Adovasio et al., 2016; Alesina et al., 2013; Destro-Bisol et al., 2004).

Even after this societal transition, the shift towards cultural norms favouring intergroup inequality took many generations to unfold (Kerig et al., 2025). Other primate species organise along much more equal gender lines when male competition is not beneficial for gaining status or rank (Arseneau-Robar et al., 2016; Surbeck et al., 2025) and in some species females engage in coalitionary aggression (Muñoz-Reyes et al., 2023; Stockley & Campbell, 2013), suggesting that the male-exclusion link is itself determined by local context and custom. Hence, gender-based hierarchies and differences in aggression are not purely based on the biological aspects of gender but emerge in concert with contextual and culturally specific factors. Given this, we might expect that such cultural factors are communicated to young children and in part contribute to the early emergence of gender-based differences in moral exclusiveness during middle childhood.

Limitations and Future Directions

There are limitations to consider in interpreting our findings, as well as several areas for future research to consider. So far, there has been only very limited study of younger children's own preferences for hierarchy and status (Cadamuro et al., 2022; Lei et al., 2025),

and we were not able to measure this in our younger childhood samples. Relatedly, we did not measure moral exclusiveness towards human outgroups amongst our very young participants. Measuring moral exclusiveness towards different outgroups as well as a burgeoning preference for hierarchies amongst a younger sample will provide an important test of our proposition that boys and girls begin to diverge at least in part due to different preferences for hierarchy. Further, our analysis of the association between preference for hierarchies and moral exclusiveness judgments in adolescence and adulthood carries the caveat of being cross-sectional. We encourage future longitudinal research in this area to better establish the predictive relationship between these factors.

Additionally, we conducted these studies within only one cultural context. In general, given the patriarchal organisation of most societies, we would expect limited cross-cultural differences in gendered moral exclusiveness. Yet, it would be a valuable next step to examine children across cultures which vary in terms of the *degree of* hierarchical structure to understand how socialisation in such cultures informs the age at which gender groups begin to diverge. Additionally, we would expect the targets of moral exclusiveness to vary by culture, even if the pattern of exclusiveness remains consistent. Exploring such differences would provide further evidence to support the idea of an evolutionary account which is informed by socially constructed ideas of who ought to be included by whom.

Table of Limitations

	Dimension	Assessment
Internal validity	Is the phenomenon diagnosed with experimental methods?	No. As the IV in this study is age, no random assignment was possible.
	Is the phenomenon diagnosed with longitudinal methods?	No

	Were the manipulations validated with manipulation checks, pretest data, or outcome data?	Not applicable – no experimental manipulation.
	What possible artifacts were ruled out?	Experimenters were kept constant across samples, ceiling and floor effects were excluded by choosing DVs with variation, task difficulty was adjusted to age of sample.
Statistical validity	Was the statistical power at least 80%?	In each sample we ran sensitivity analyses with statistical power set at 80%. These analyses indicated that we were able to detect our effect sizes at 80% power with our sample sizes.
	Was the reliability of the dependent measure established in this publication or elsewhere in the literature?	Paper includes reliability analysis and measures have been previously used reliably with different samples in published literature
	If covariates are used, have the researchers ensured they are not affected by the experimental manipulation before including them in comparisons across experimental groups?	Not applicable
	Were the distributional properties of the variables examined and did the variables have sufficient variability to verify effects?	Yes
Generalizability to different methods	Were different experimental manipulations used?	No
Generalizability to field settings	Was the phenomenon assessed in a field setting?	No

	Are the methods artificial?	Yes, completing a survey in a classroom or home setting can be considered artificial
Generalizability to times and populations	Are the results generalizable to different years and historic periods?	Not tested – likely that moral expansiveness has changed across historic periods, but evidence points to men being more morally exclusive than women at least since the period after homo sapiens began living in sedentary settlements.
	Are the results generalizable across populations (e.g., different ages, cultures, or nationalities)?	Our paper demonstrates age differences. In terms of cultural differences, this remains an open question, although literature does point to gender differences in moral exclusiveness across cultures.
Theoretical limitations	What are the main theoretical limitations?	While we make theoretical claims about gender differences in moral exclusiveness between children, adolescents, and adults, we measure these effects in one cultural context, using cross-sectional data. The strongest evidence for this theoretical position would require longitudinal data collected across cultures, so we suggest caution in interpreting our findings.

Conclusion

Assessing participants aged 4 to 80, we document a gender difference in moral exclusiveness that manifests towards different human and non-human groups: males are more morally exclusive than females. This difference emerges most strongly in adolescence and persists among adults. Importantly, instances of this gender difference are already apparent

among 6-year-old children. Together this evidence supports the idea that we are socially prepared for gender role differences in our attitudes towards hierarchies and dominance in childhood, with further differences between male and female adolescents brought out between 13- and 18-years. These findings centre childhood as a period of great importance for those seeking to challenge gender-based notions of moral exclusiveness.

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